



Mingle

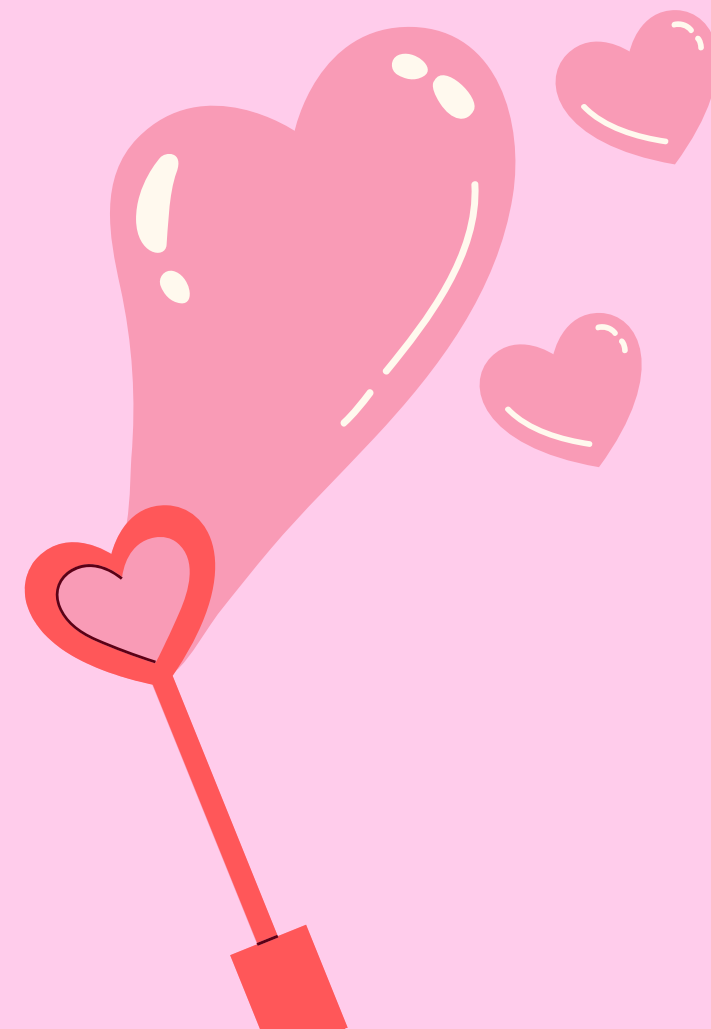
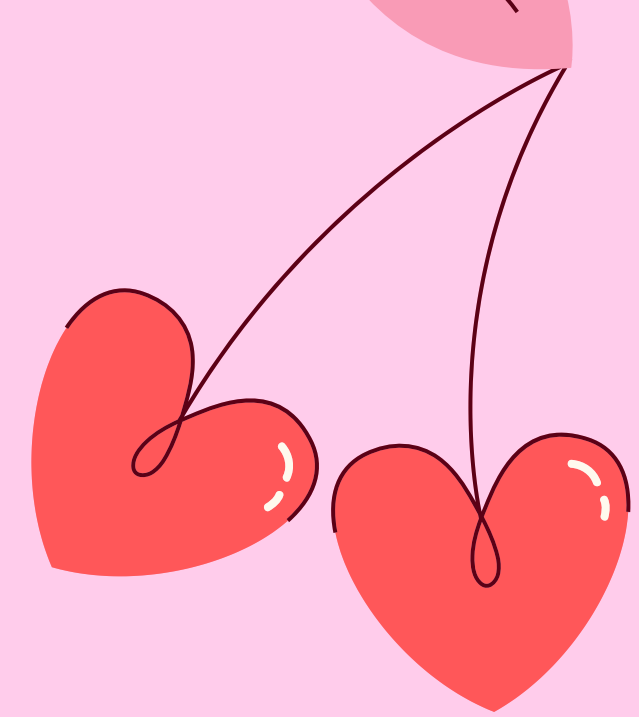
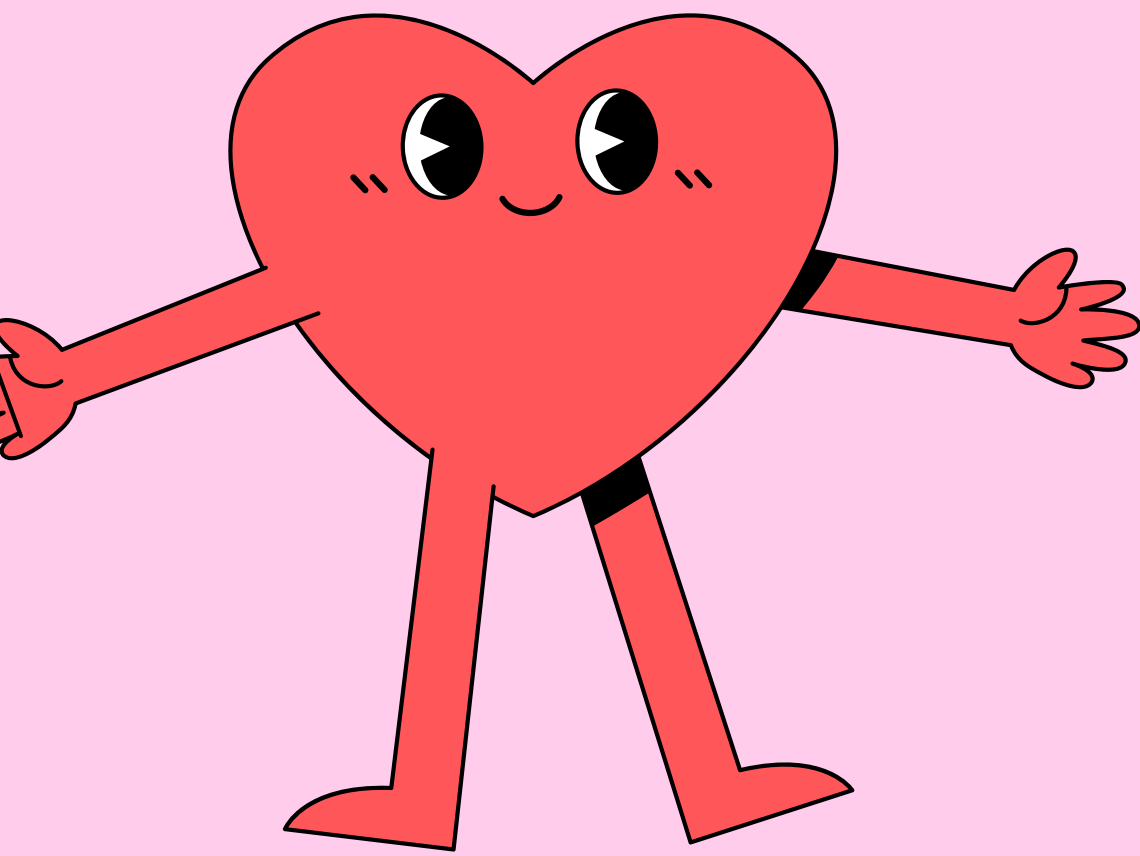
Dating app

Hadi JABER

Mohamad Hassan IBRAHIM

Karen EL KHOURY

Malek TURKI



What is Mingle?



Making online interactions more natural and meaningful

A geo-social dating application

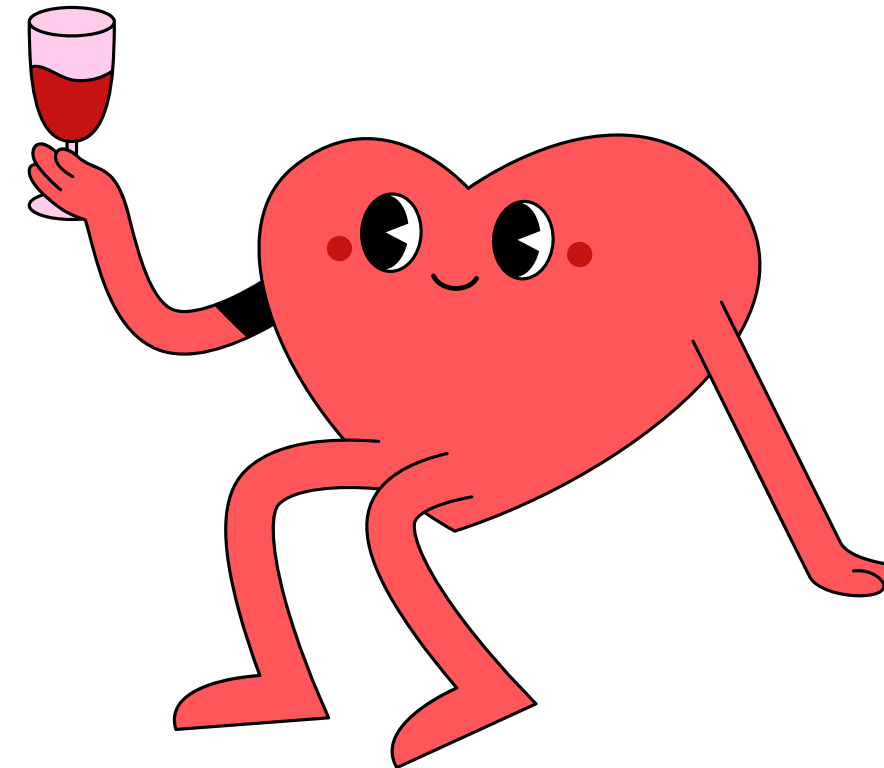
Helps users initiate and maintain conversations

Combines:

Location-based discovery

AI conversation assistant

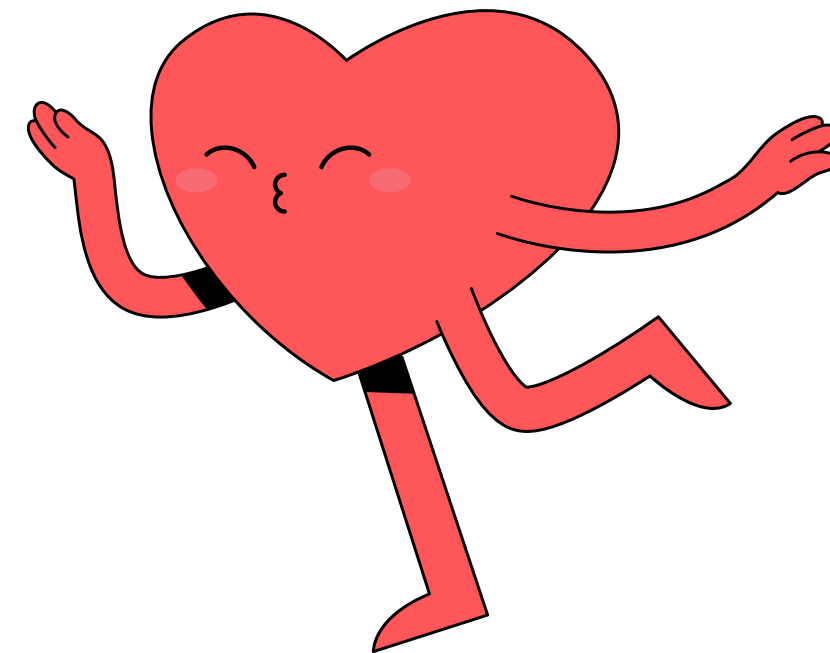
Adaptive chat interface

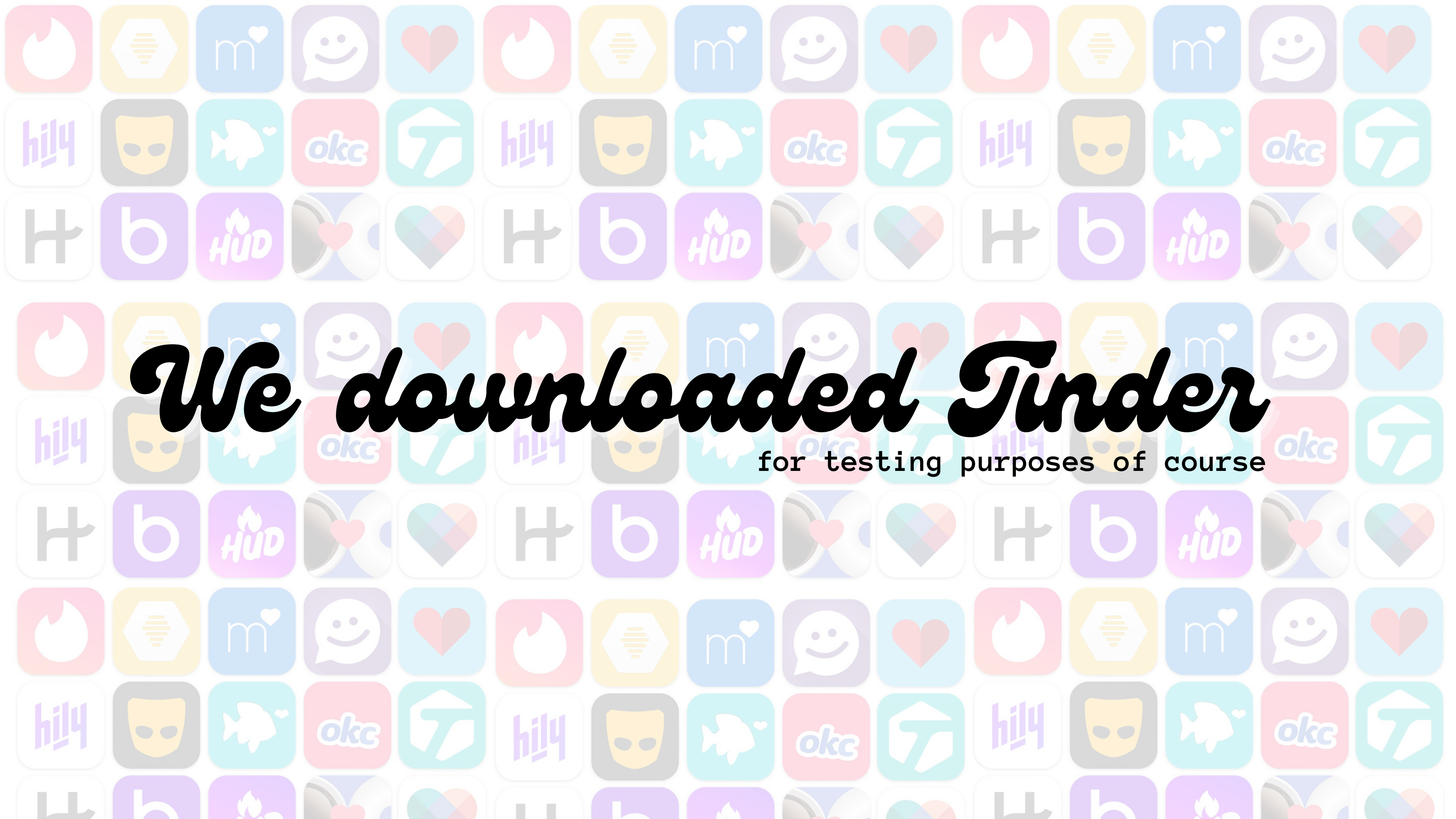


Problem Identification



- 1- Testing Tinder/Pure/Hinge
- 2- Survey Results





We downloaded Tinder

for testing purposes of course

We downloaded Tinder for testing purposes of course

What we liked: Intuitive UI

Great Color Set

Easy to Use

What we didn't like:

Doesn't help in finding someone specific

Some of us couldn't send the first message because we were too shy

We also tried the Pure Dating app (BDSM)

What we liked:

The UI is very creative (TOO MUCH)

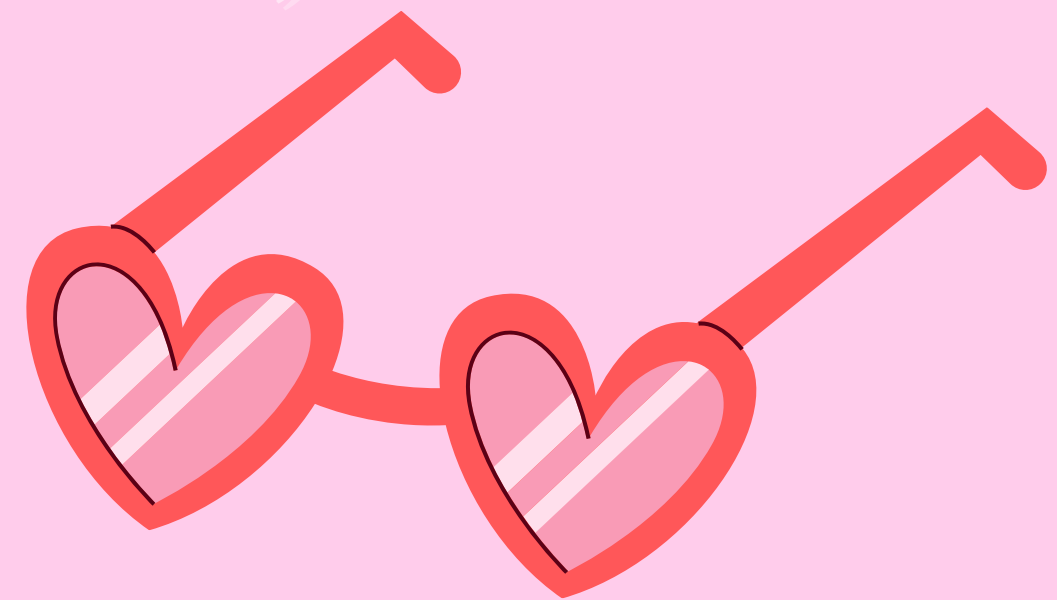
Offers an interesting option of "online relation"

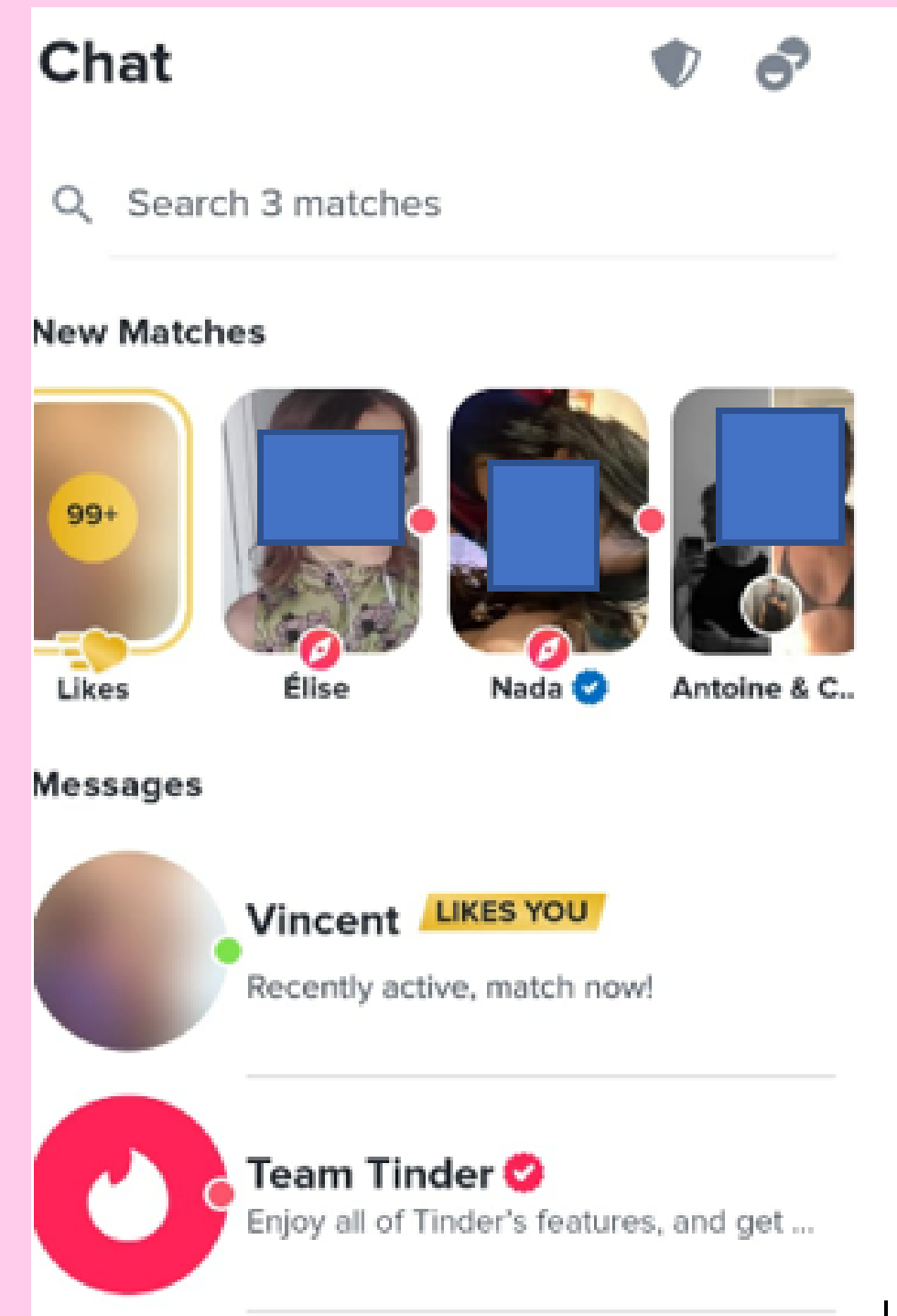
What we didn't like:

The UI is overstimulating! Drawings everywhere

Took us more than 10 min to create our profile

We didn't understand how the app works, how to match, and uninstalled the app real quick

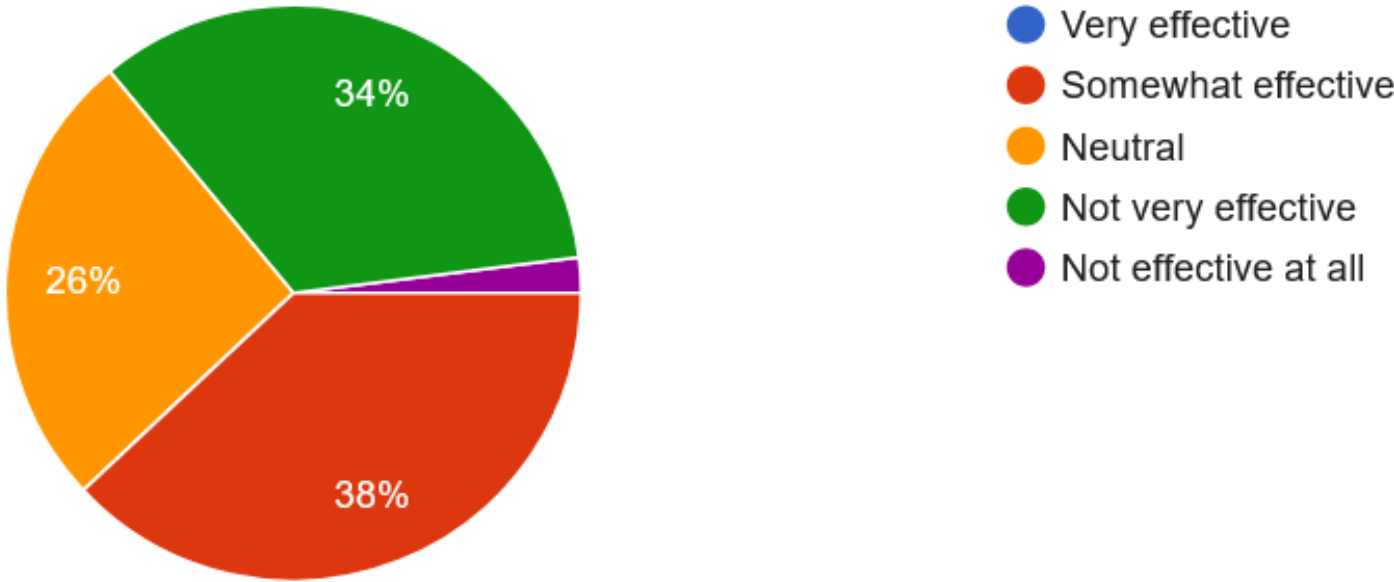




Do you think these apps are effective for meeting new people?

[Copier le graphique](#)

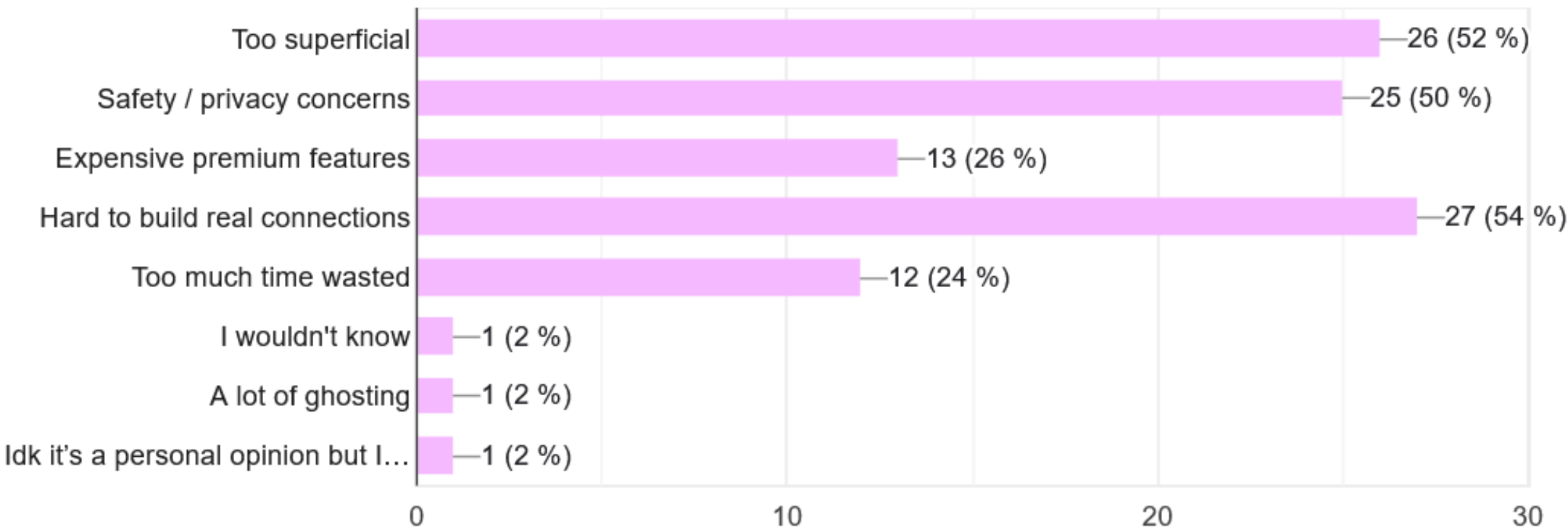
50 réponses



What do you think are the biggest disadvantages?

[Copier le graphique](#)

50 réponses



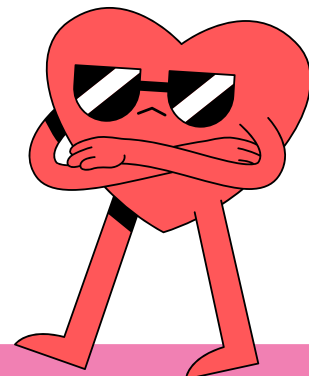
User persona



Alex (Introvert)

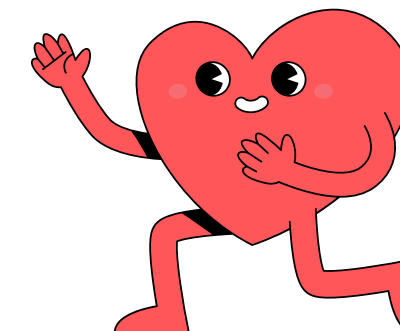
- Struggles to start conversations
- Overthinks messages
- Uses dating apps occasionally

==> Chats often stay silent



Maria (crushing on her classmate)

- She sees someone on campus
- But cant approach him/her nor knows his/her name to find them on social media



Usability criteria

Criterion	Success Metric
Learnability	Profile created no help requested
Efficiency	Time-to-first-message reduced vs. Tinder baseline (no AI suggestions)
User Control	100% of users can send a message without using any AI suggestion; 0 users feel "forced"
Non-Intrusiveness	Flow Mode correctly suppresses suggestions;
Accessibility	No critical action relies solely on red/green colour coding; both themes pass contrast check

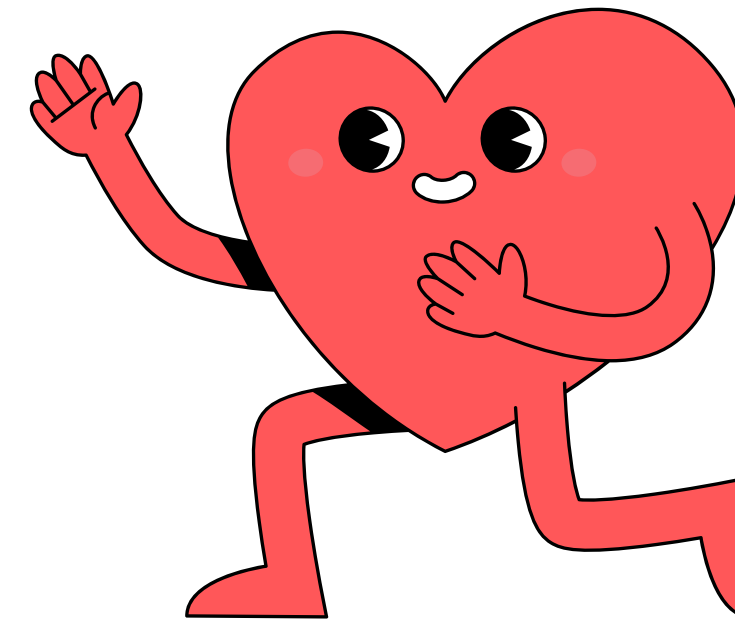
Design Solution



Main Tasks Supported

Users can:

1. Discover nearby users
2. Match with profiles
3. Start conversations
4. Continue or revive chats
5. Manage their profile



Design Solution



AI Assistant

suggests conversation starters

Adaptive Chat

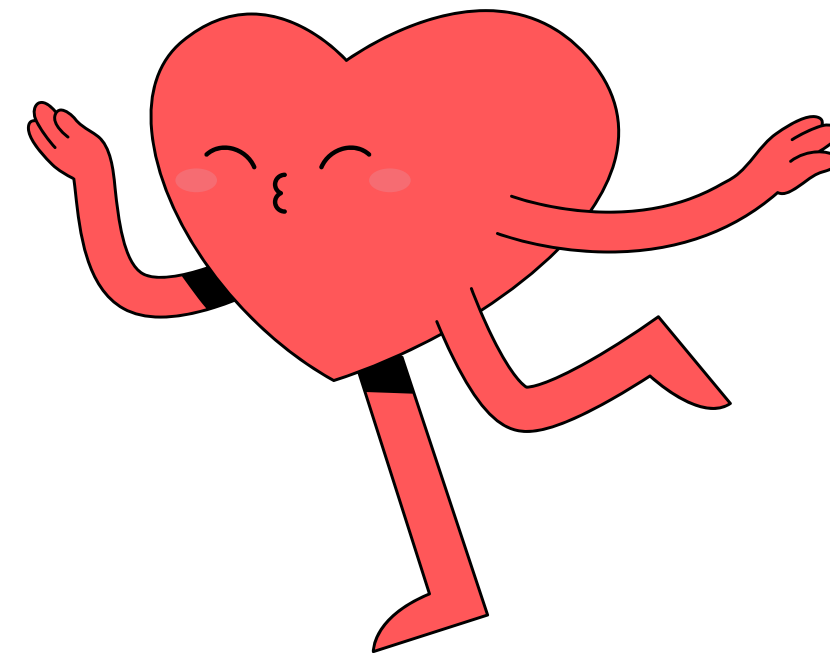
detects inactivity and suggests topics

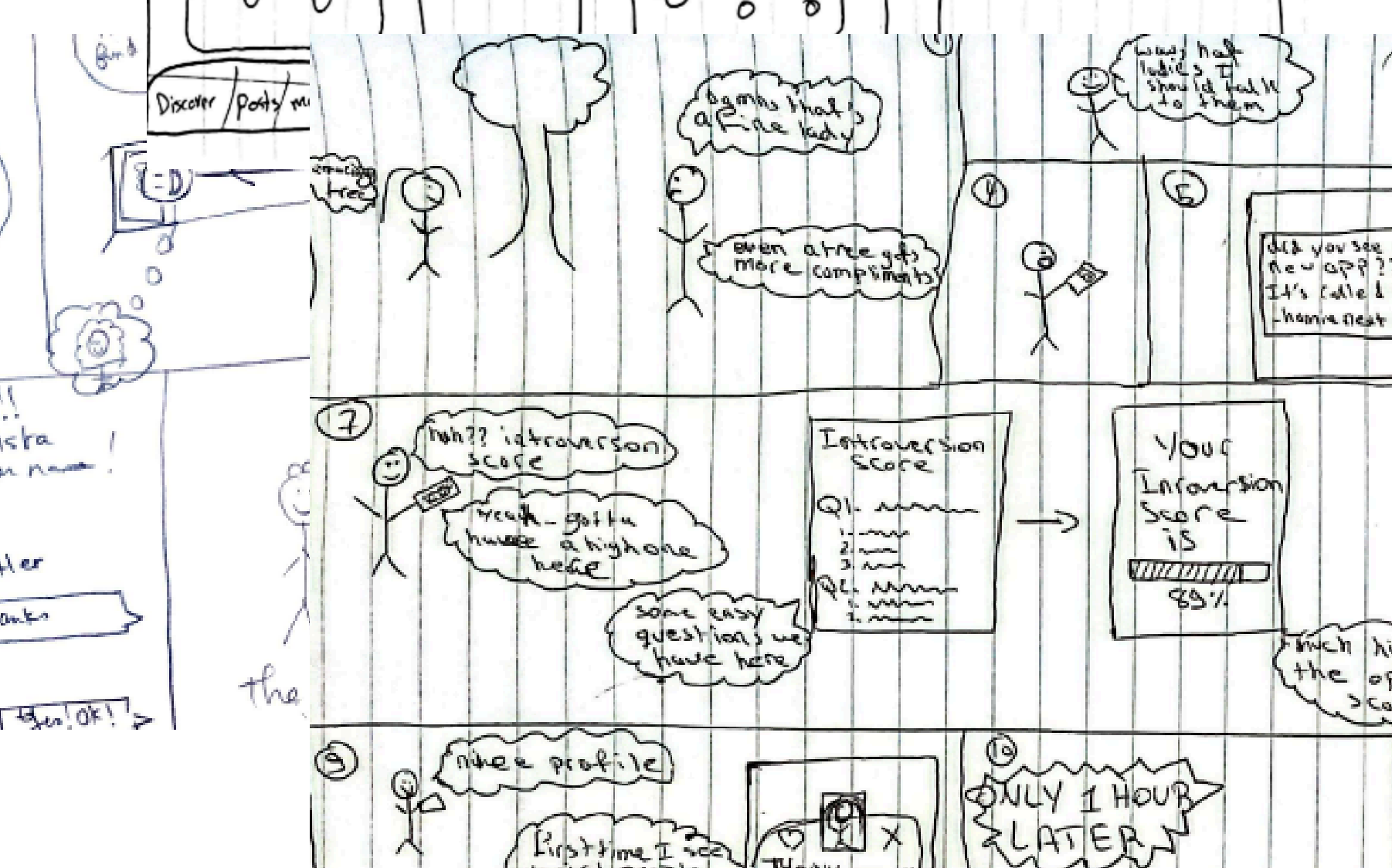
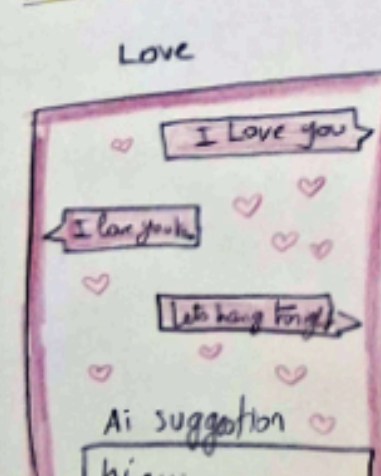
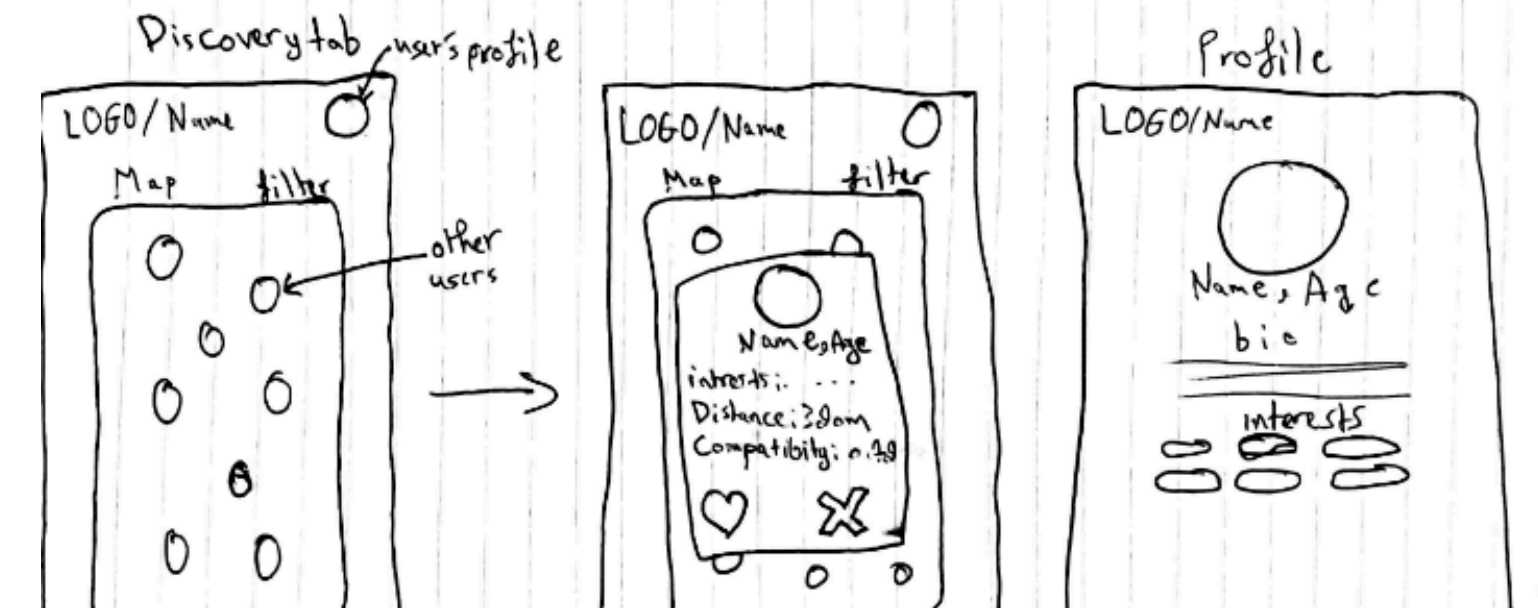
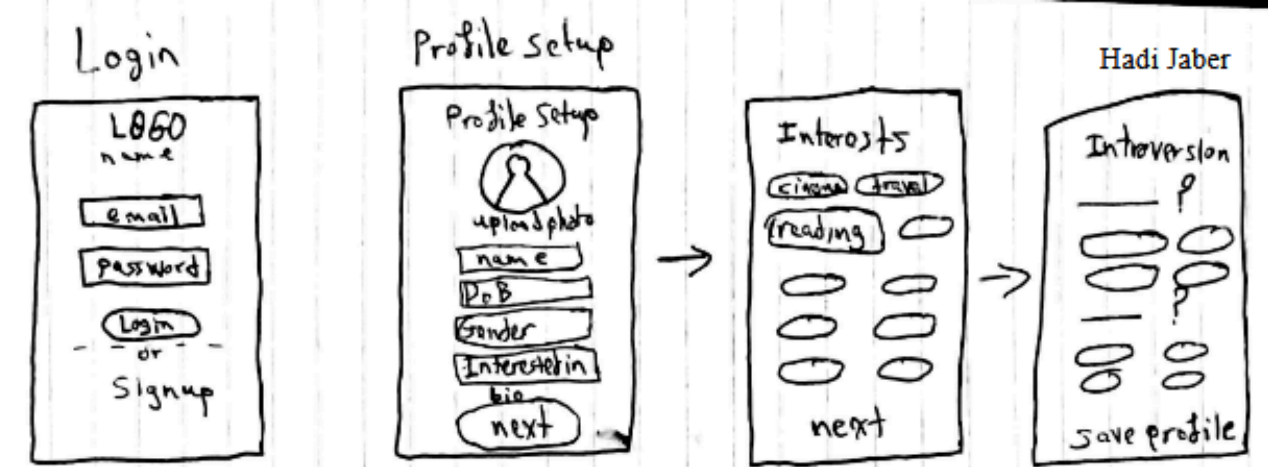
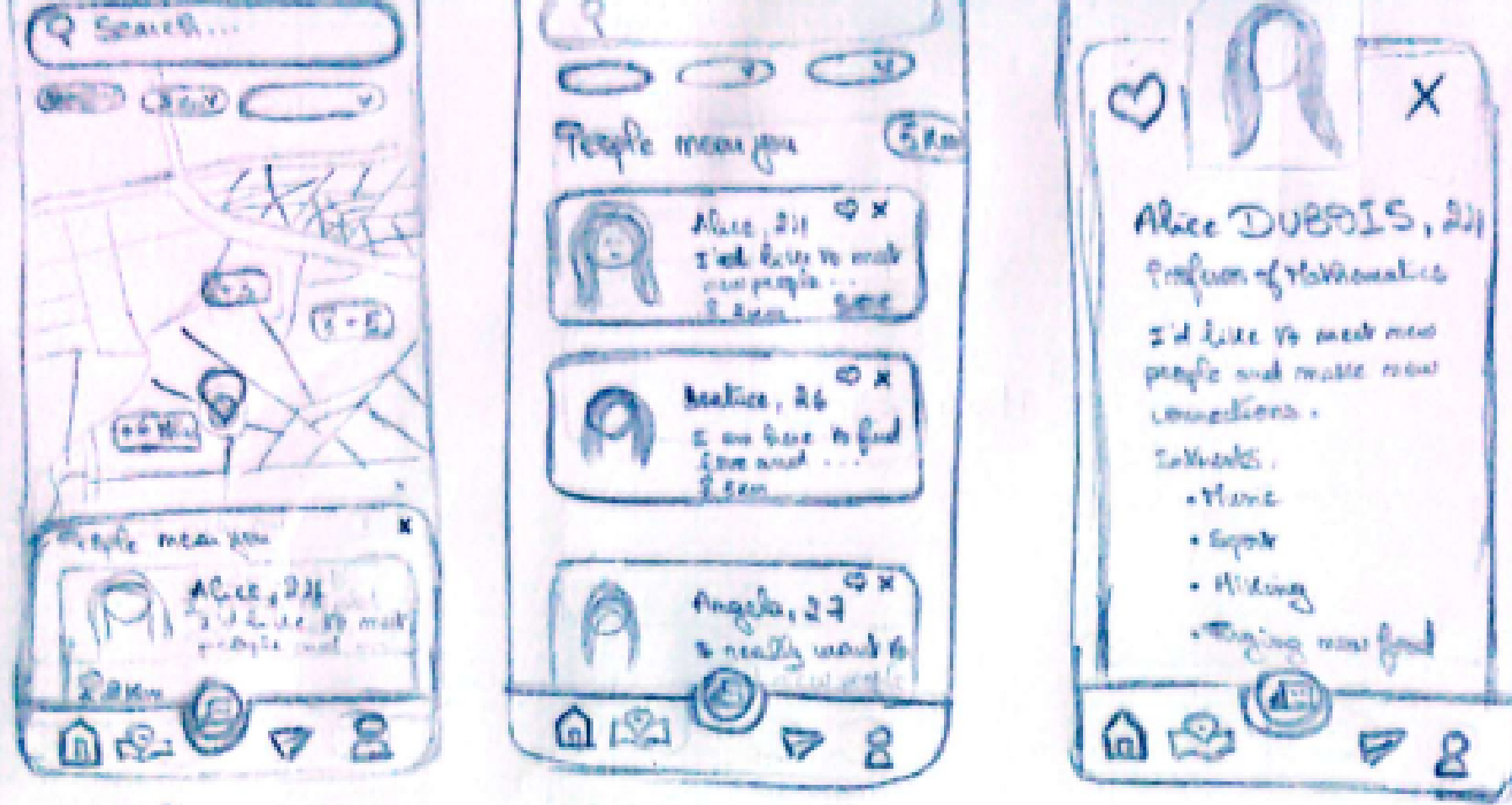
Geo-social Map

discover nearby users

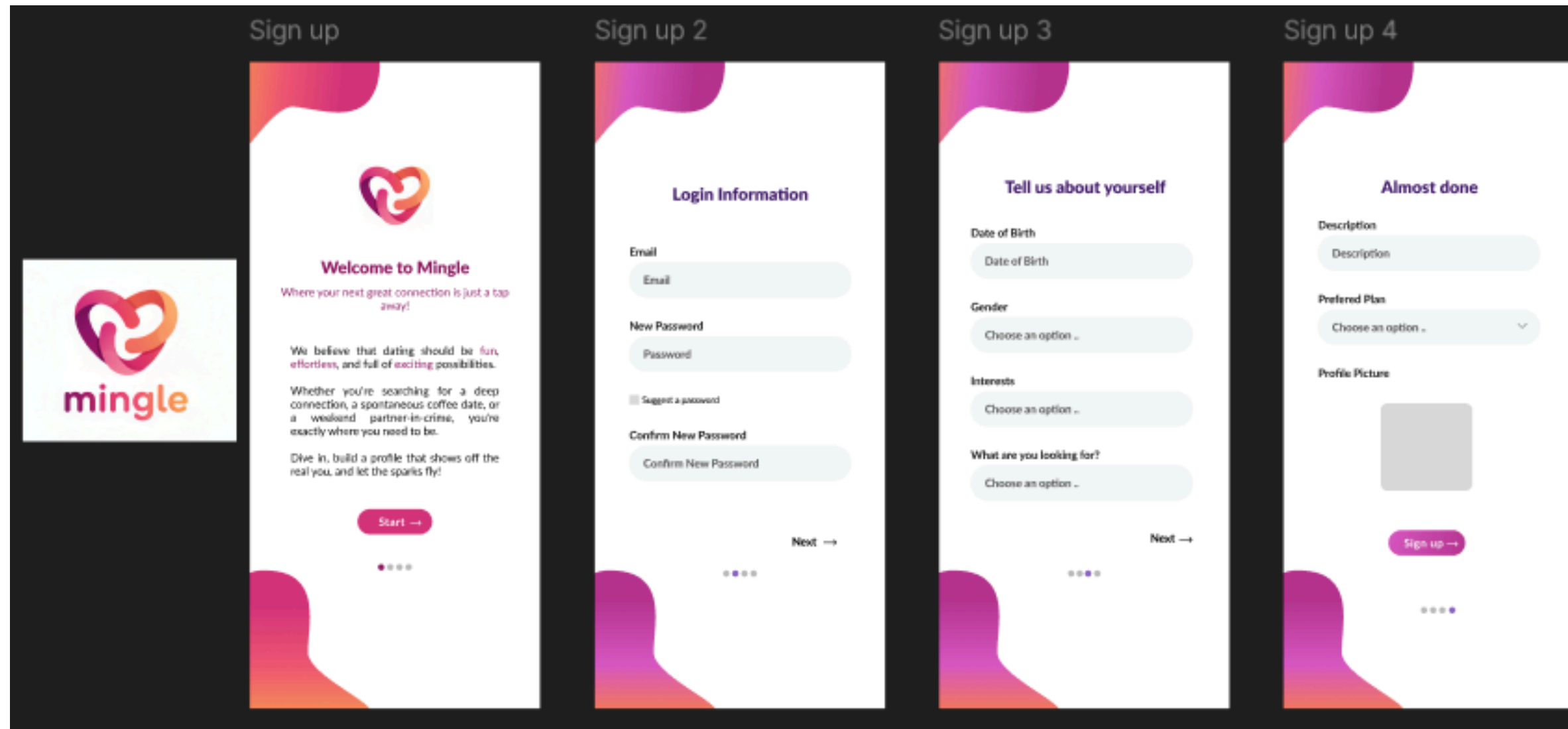
Progressive Onboarding

reduces user fatigue

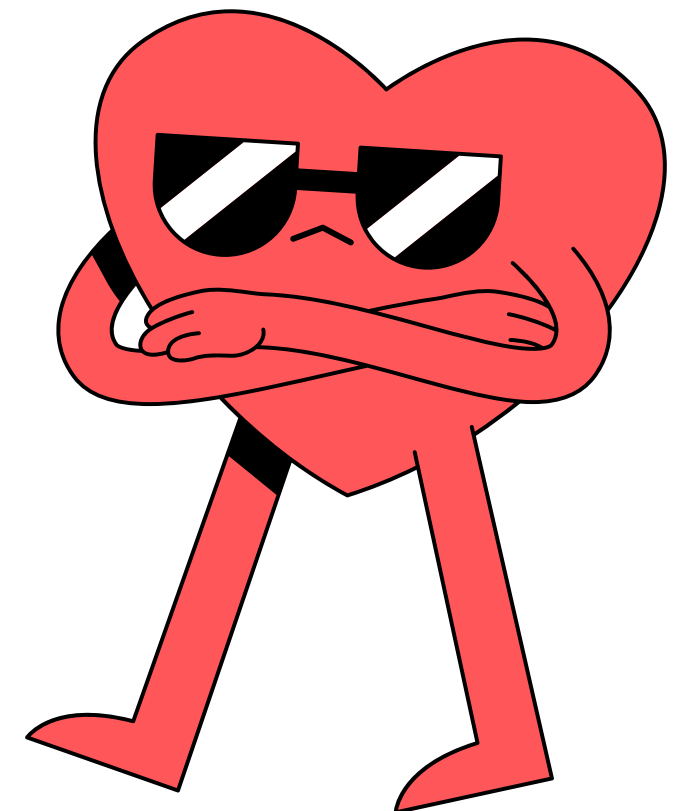




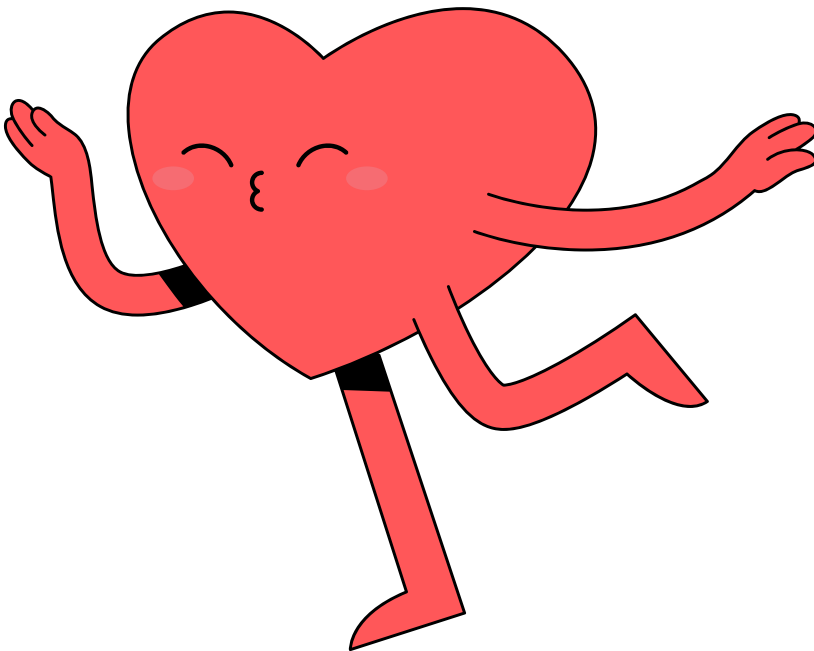
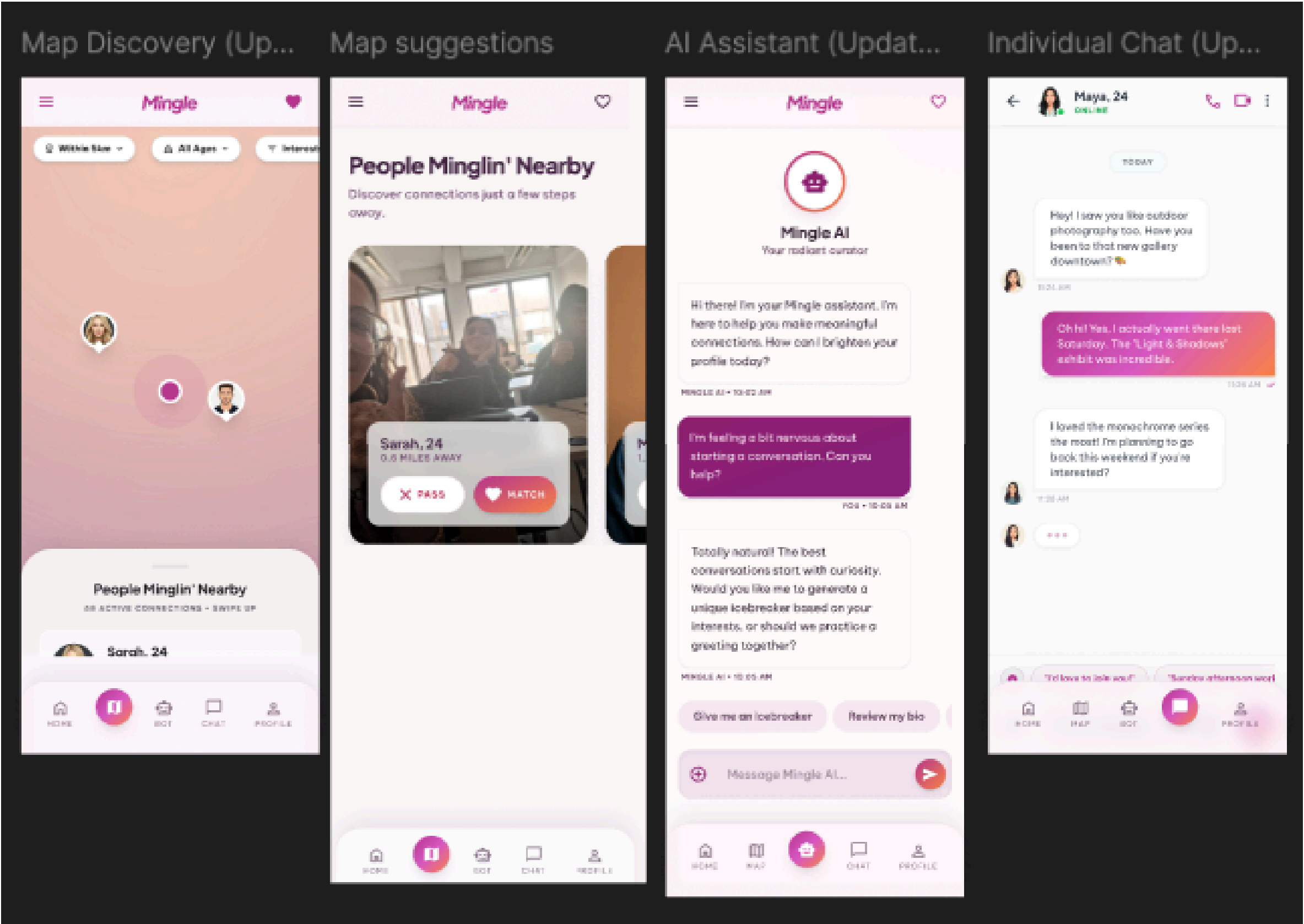
Designing the UI – figma



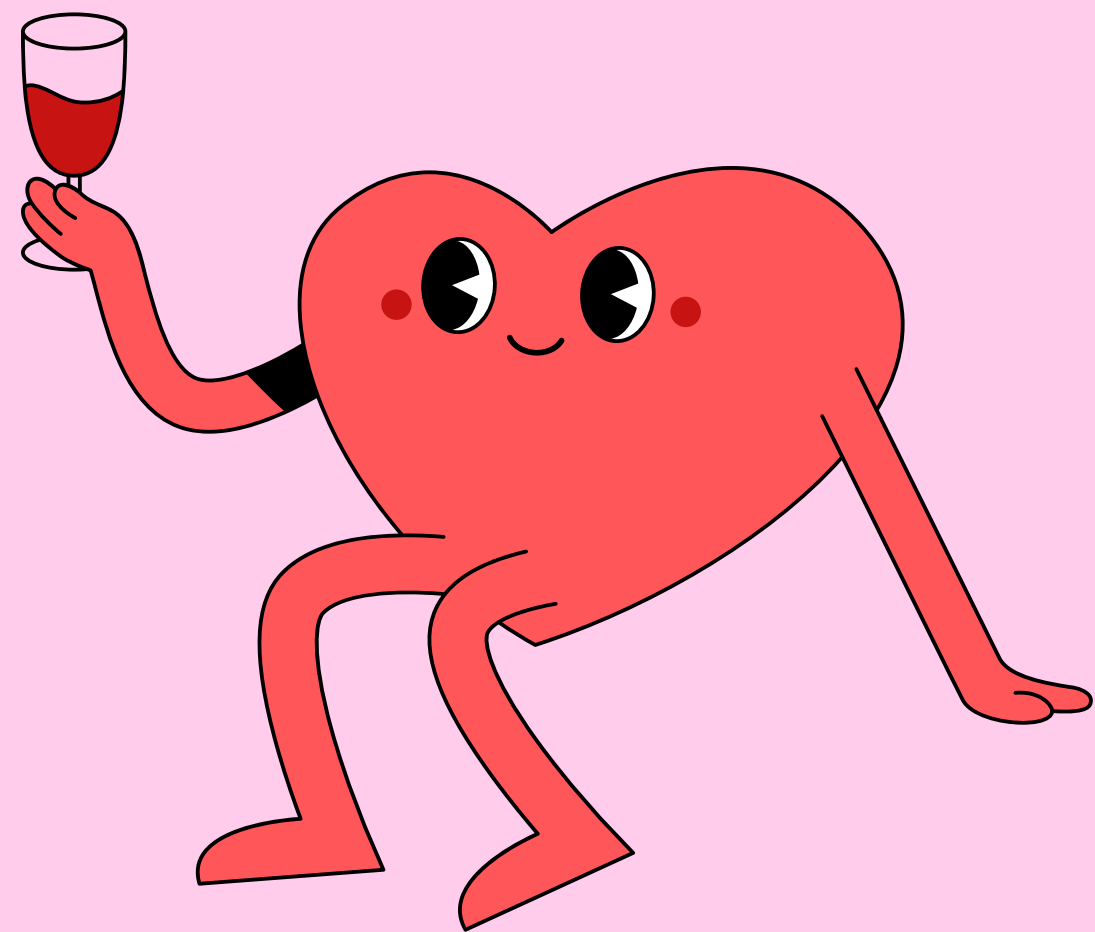
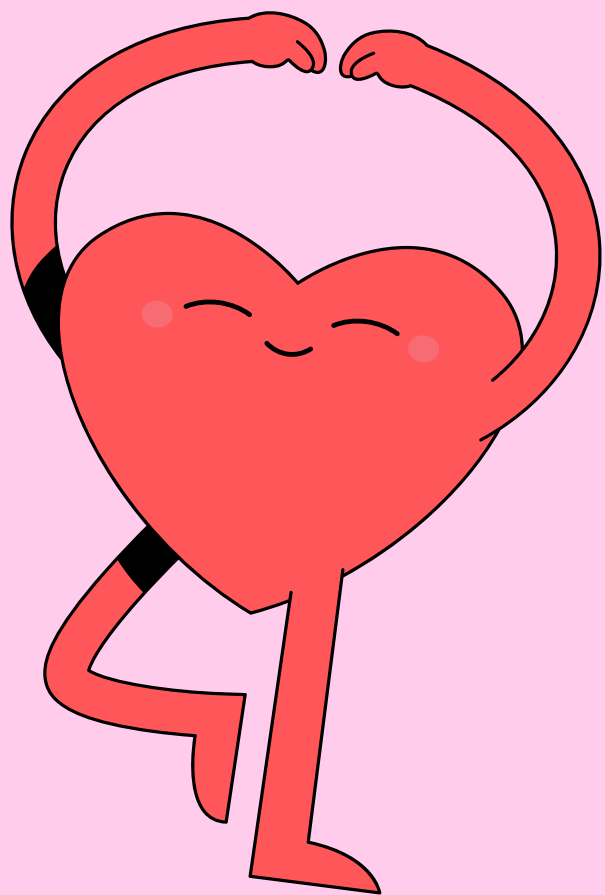
Fitts' Law
3-Click Rule
Grouping (Gestalt Principle)



Designing the UI – figma



Impementation



Architecture

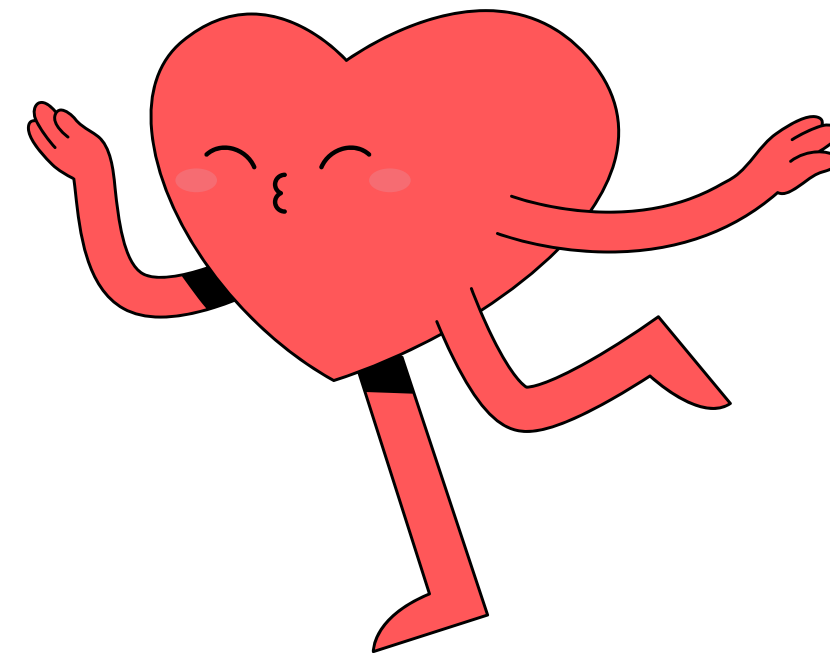


System Architecture

- Mobile app (React Native + Expo)
- Backend (AWS serverless architecture)
- REST API communication

Technologies:

- Frontend: React Native, Expo Router
- Backend: Lambda, API Gateway
- Database: PostgreSQL + PostGIS
- Storage: S3

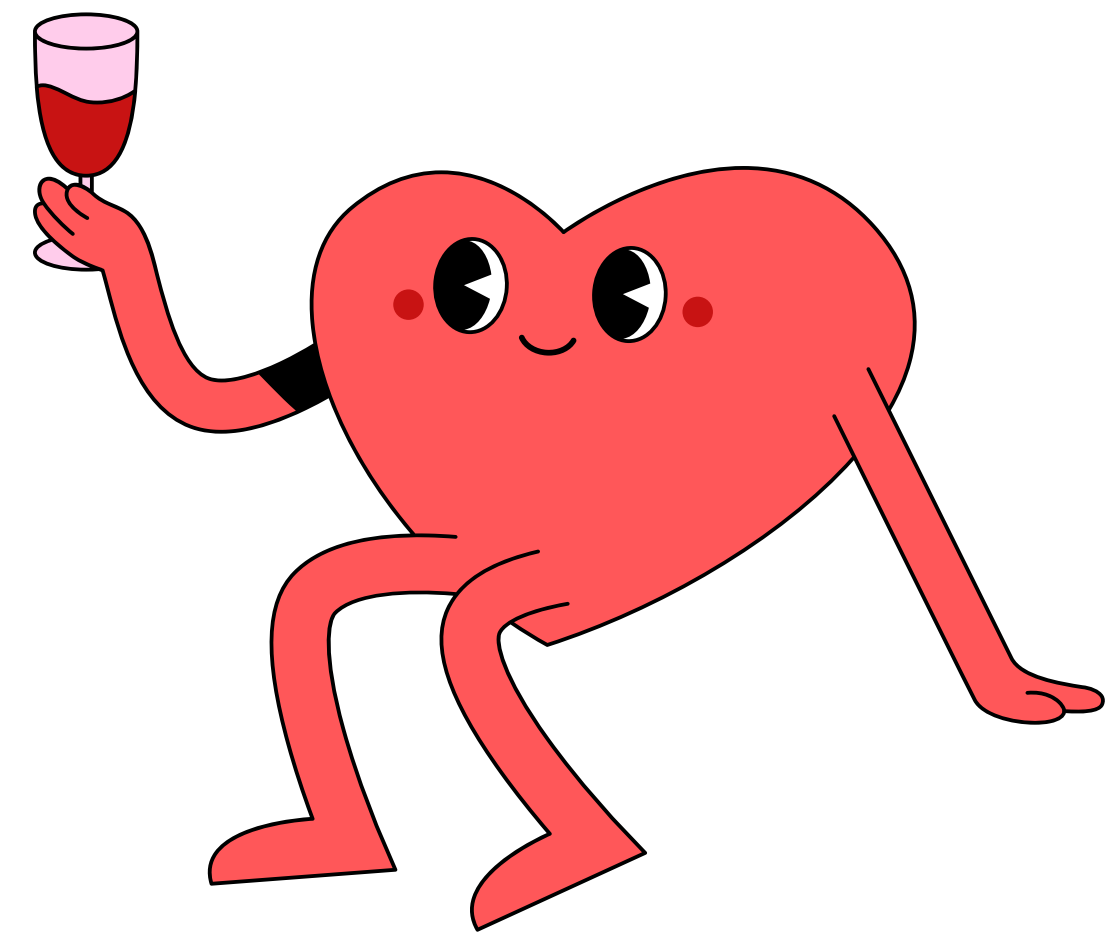


Core Features Implementation



Core Features

- User profiles & authentication
- Geo-social discovery (map + nearby users)
- Matching system (likes → matches)
- Real-time chat system
- Social features (posts, comments, stories)



Core Features Implementation



Matching with Introversion Score

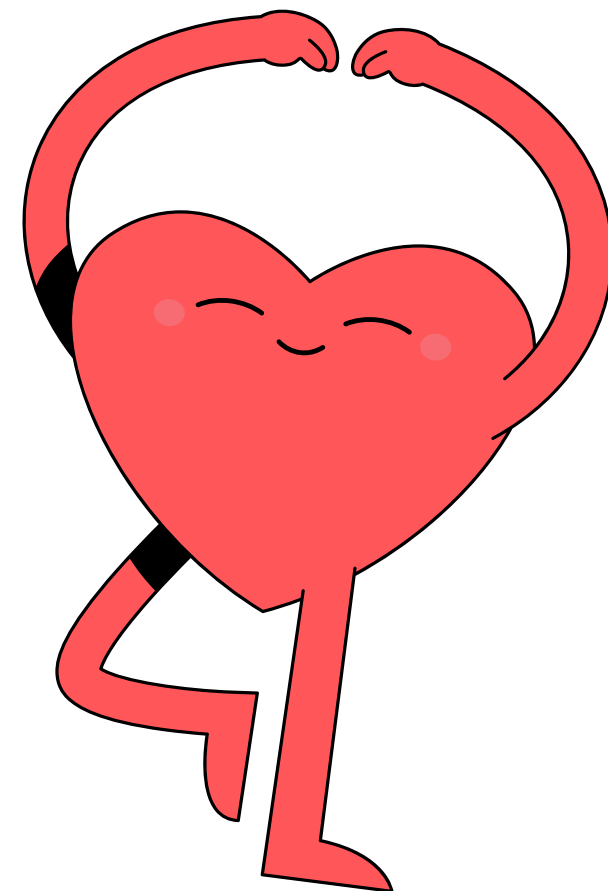
Compatibility Score

$\text{Score} = 0.5 \cdot \text{interests} + 0.3 \cdot \text{distance} + 0.2 \cdot \text{introversion}$ How Introversion Similarity is Calculated

- Each user has an introversion score (between 0 and 1)
-

Similarity between Users

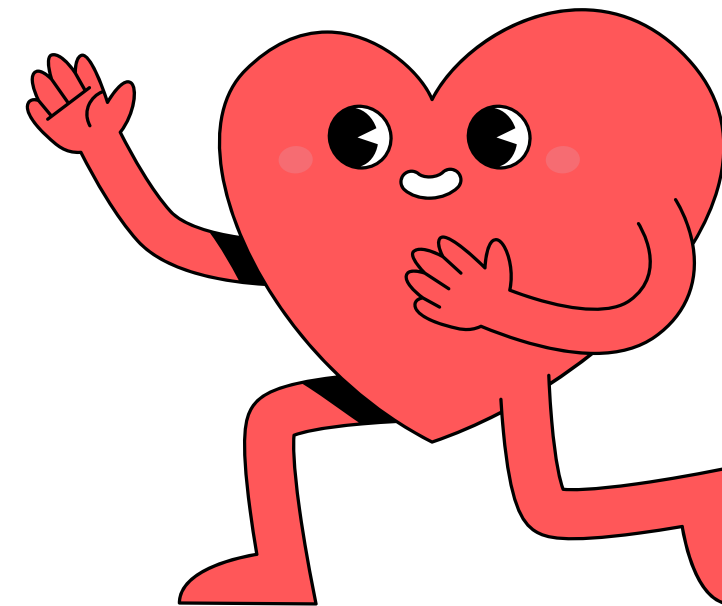
$\text{introversion_similarity} = 1 - | \text{user_1_score} - \text{user_2_score} |$

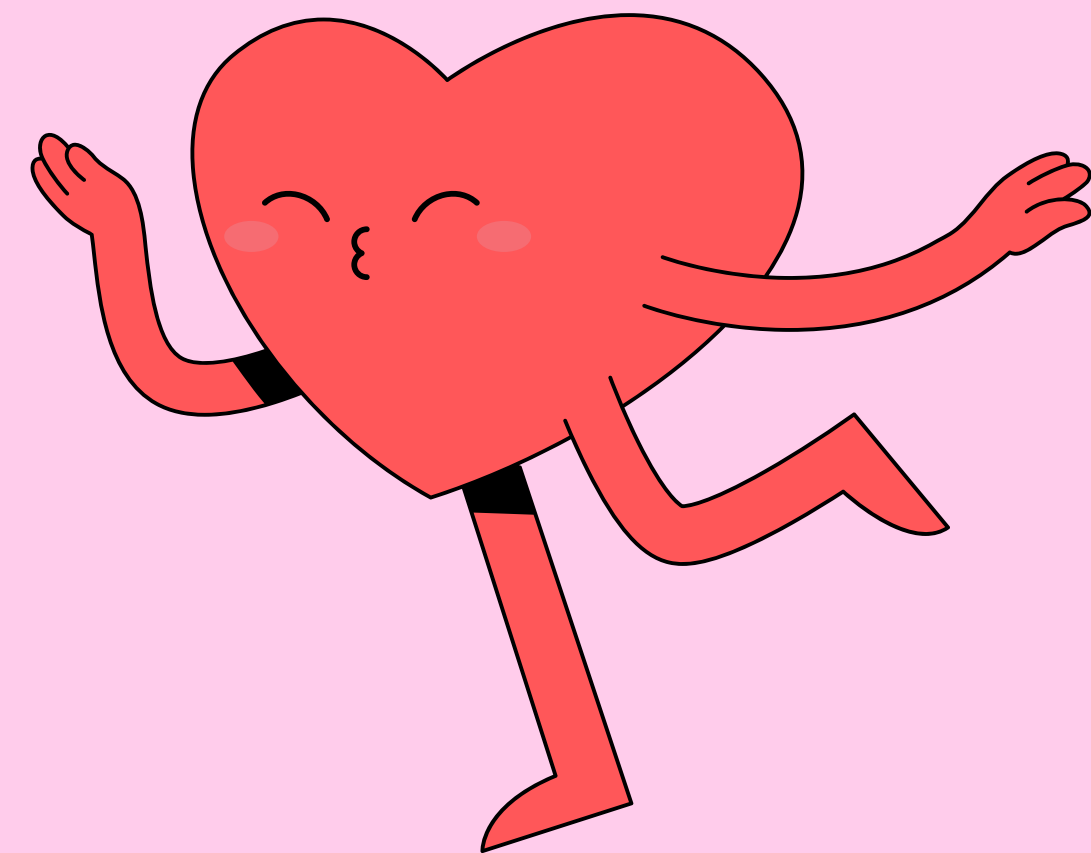
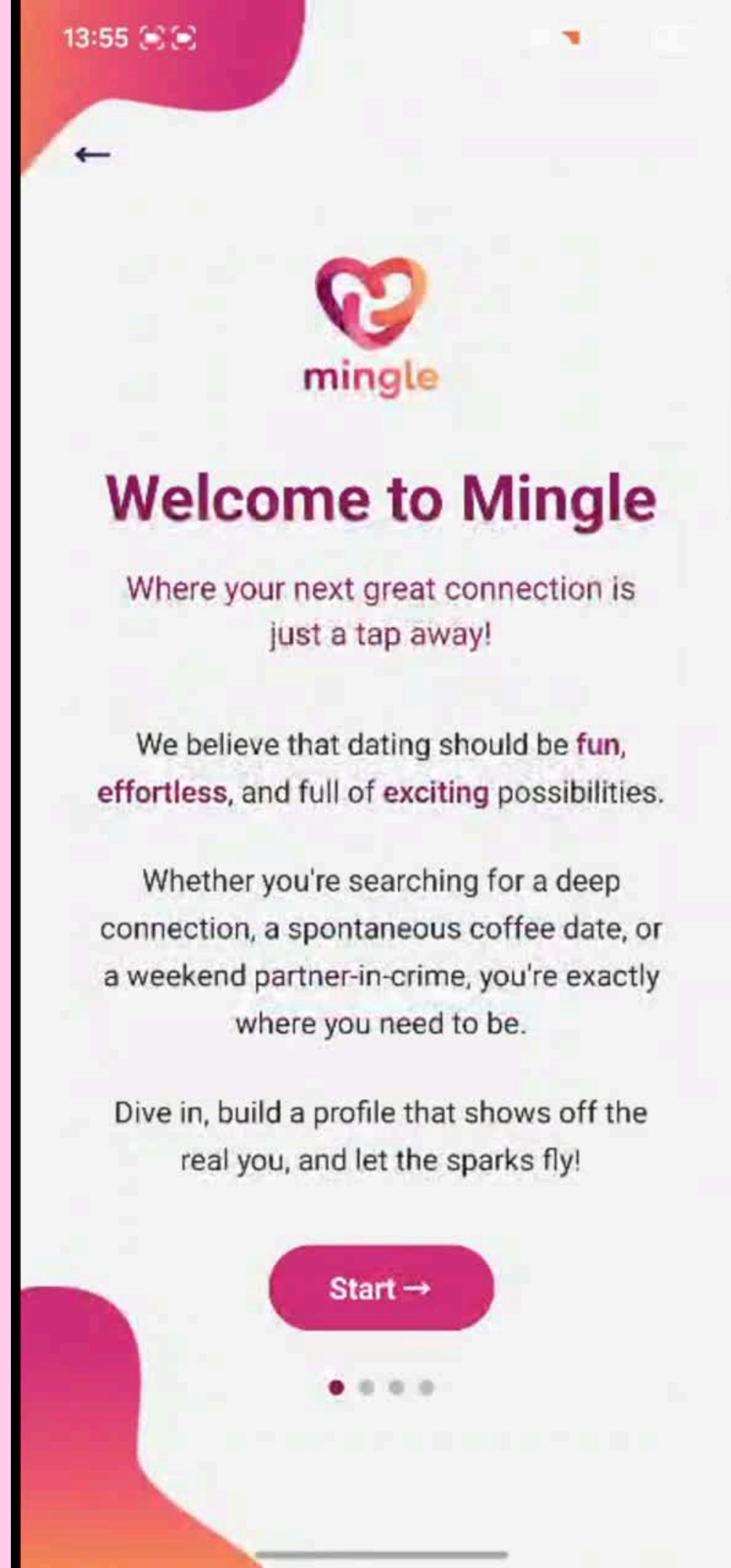


AI & UX Implementation

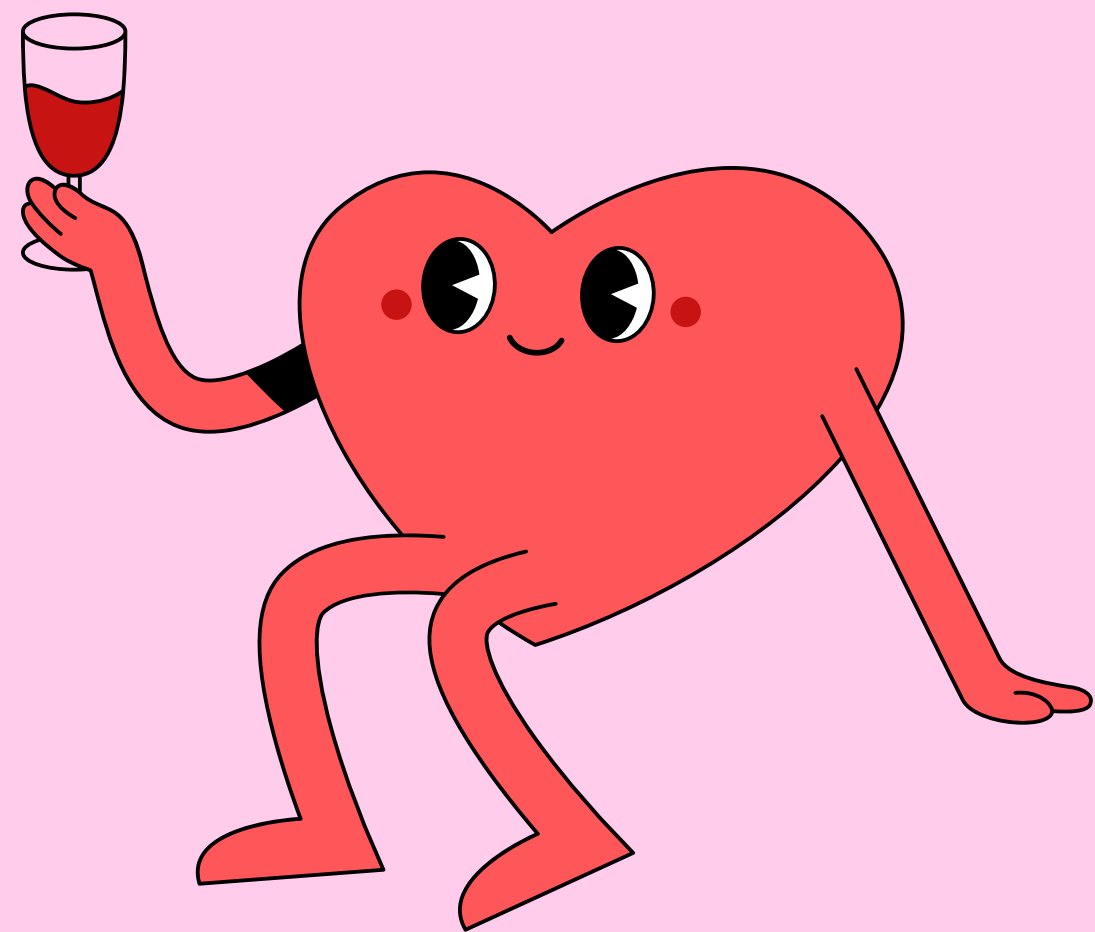
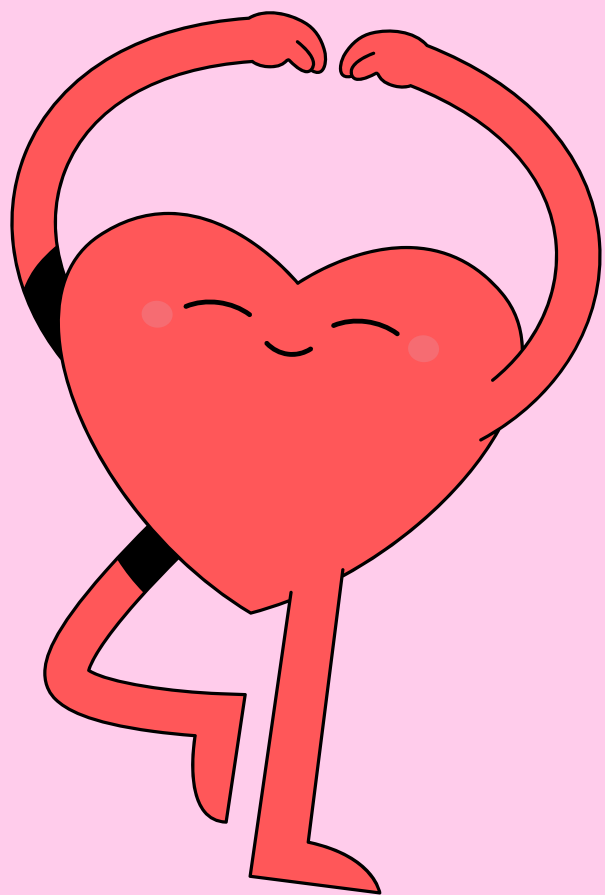


- AI suggestions
generate replies based on user input
- AI assistant
integrated in chat interface
- Conversation flow
dynamic replies + state-based interaction
- Frontend UX
reusable components (posts, stories)
progressive onboarding
input validation & user feedback





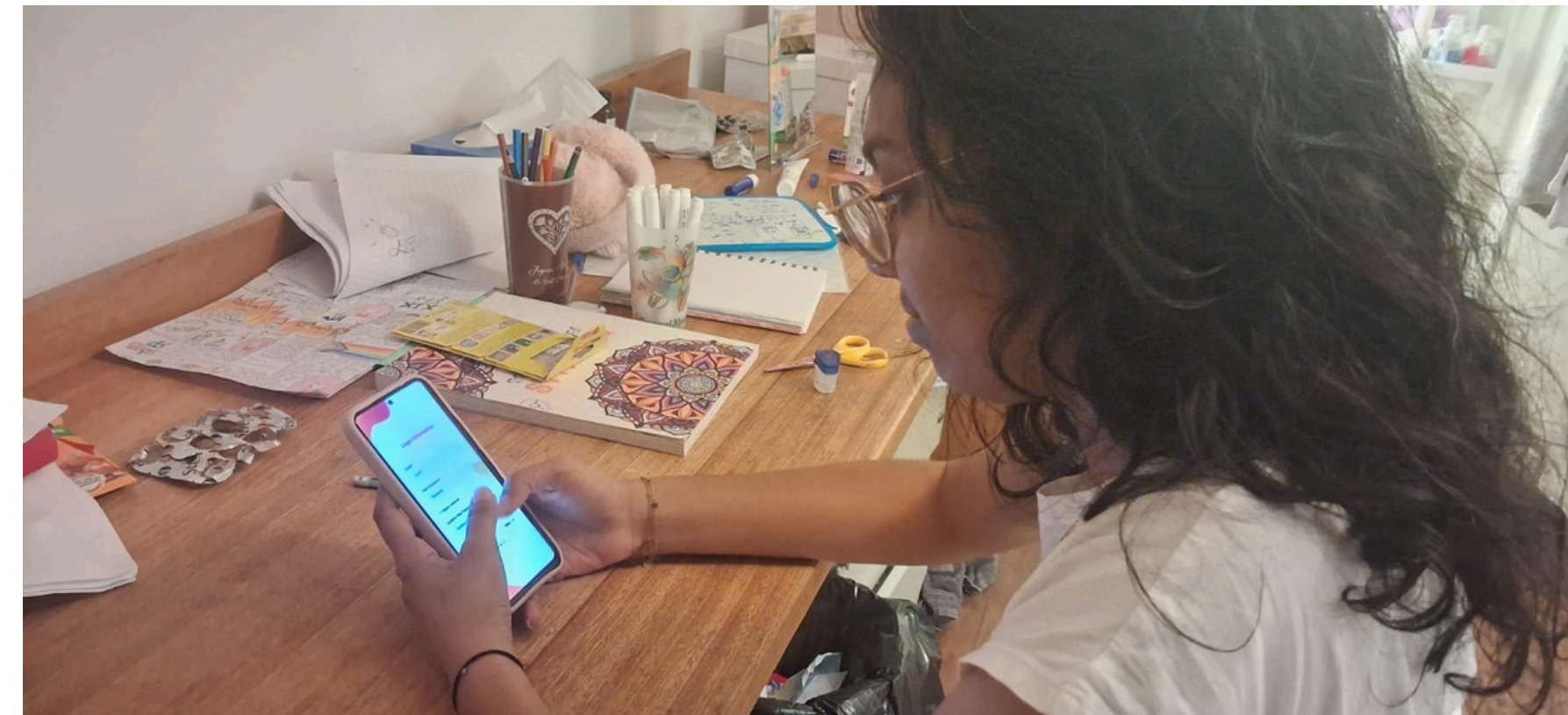
Evaluation Methods



Evaluation Methods



- We used task-based usability testing + think-aloud
- 5 participants (students: introverts, ambiverts, extrovert)
- Sessions ~10 minutes
- Users explained what they were thinking while using the app
- Answered a questionnaire after evaluation



Evaluation Methods



Tasks performed by participants

- Create a profile (onboarding)
- Discover & match using the map
- Start a conversation with AI

Results & Insights

What worked well

- Easy onboarding and navigation
- Users understood the interface quickly
- AI helped users start conversations
- Users liked having control over AI

What can be improved

- AI suggestions not always perfect
- Some hesitation to use AI
- Privacy concerns (location)

Future Work



Aspect	Strengths	Weaknesses
Conversation Experience	AI assistant helps users start and maintain conversations, reducing anxiety	Risk of over-reliance on AI, making conversations feel less authentic
User Control	AI suggestions are optional, editable, and non-intrusive	Some users may still feel influenced or monitored by AI presence
Geo-Social Map	Enables real-world, nearby discovery → more natural interactions	Raises privacy and safety concerns (location sharing, visibility)
Usability & Learnability	Simple, intuitive navigation with familiar patterns (3-click principle)	Multiple features (map, chat, AI, posts) may still overwhelm some users
Matching System	Compatibility-based algorithm (interests, distance, introversion) improves relevance	Algorithm is not fully tested → may produce inaccurate matches

Future Work

Improvements we want to make

- Improve AI suggestions to be more personalized and natural
- Improve the matching algorithm
- Add accessibility features (especially for visually impaired users)
- Strengthen privacy and security (location control, data protection)

L'accessibilité pour tous



Thank you!

Now let's go find love on Mingle

